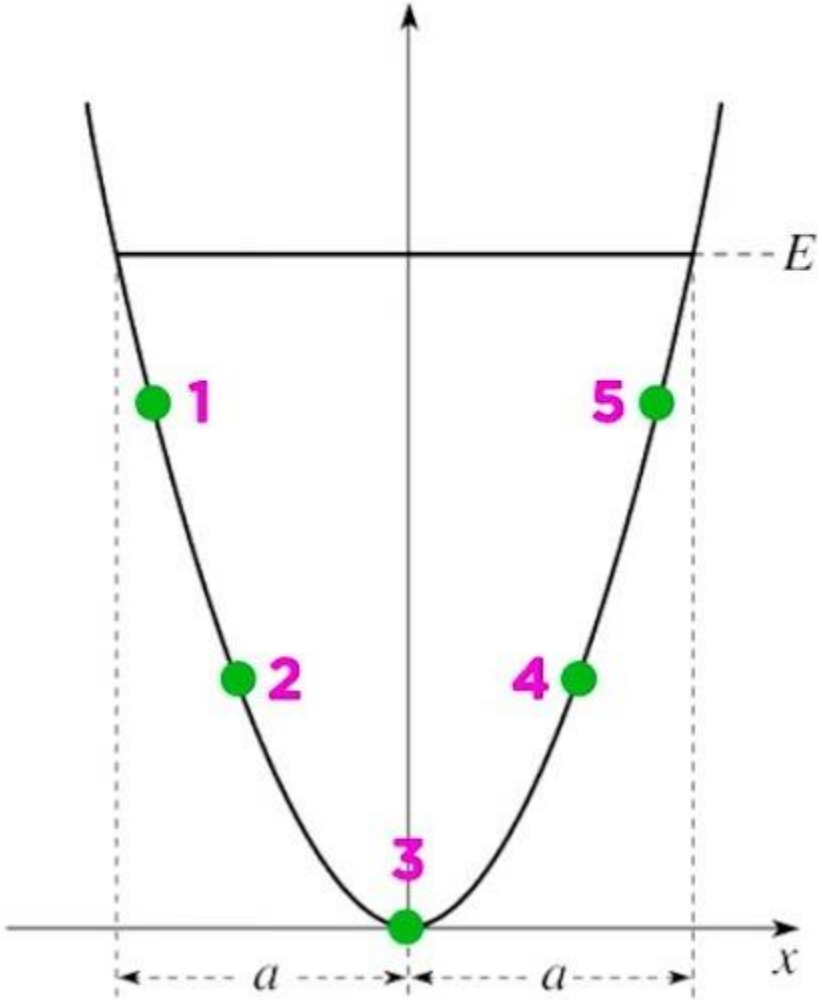
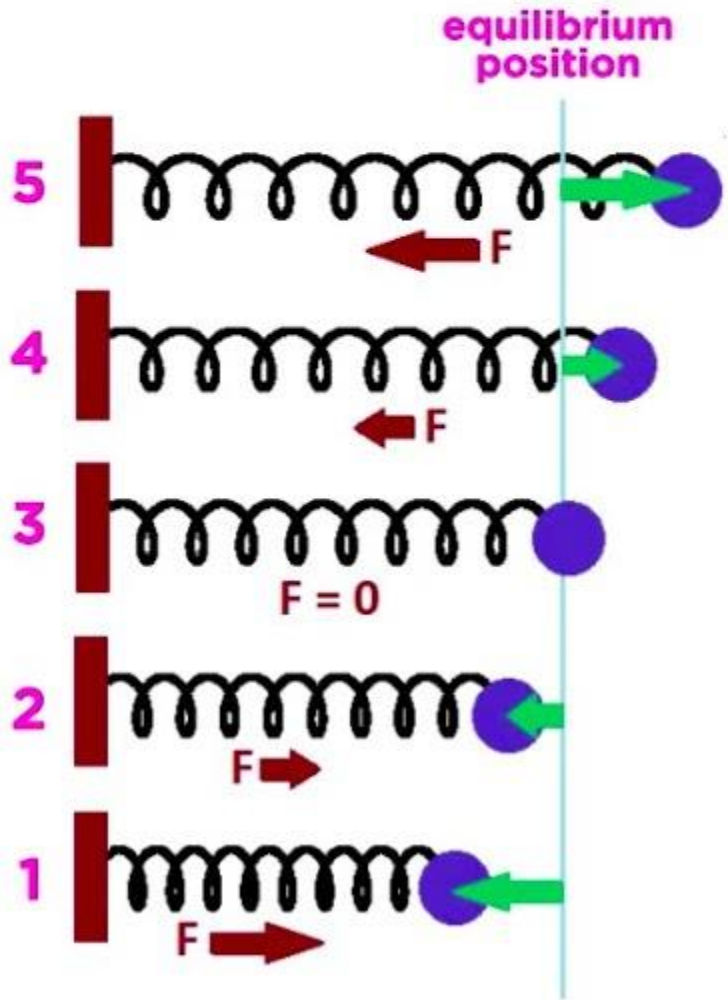
$$V(x) = \frac{1}{2}kx^2$$

expression for the classical harmonic oscillator





The Selection Rule for the Harmonic Oscillator

$$\Delta\upsilon = \pm 1$$

Vibrational energy changes will only give rise to an observable spectrum

- **if the vibration can interact with radiation**
- **if the vibration involves a change in the dipole moment of the molecule**

Vibrational spectra will be observable only in **heteronuclear diatomics** since homonuclear molecules have no dipole moment.

$$\varepsilon_{\upsilon} = \frac{E_{\upsilon}}{hc} = \left(\upsilon + \frac{1}{2}\right)\overline{\omega}_{osc} \text{ cm}^{-1}$$

Applying Selection Rule

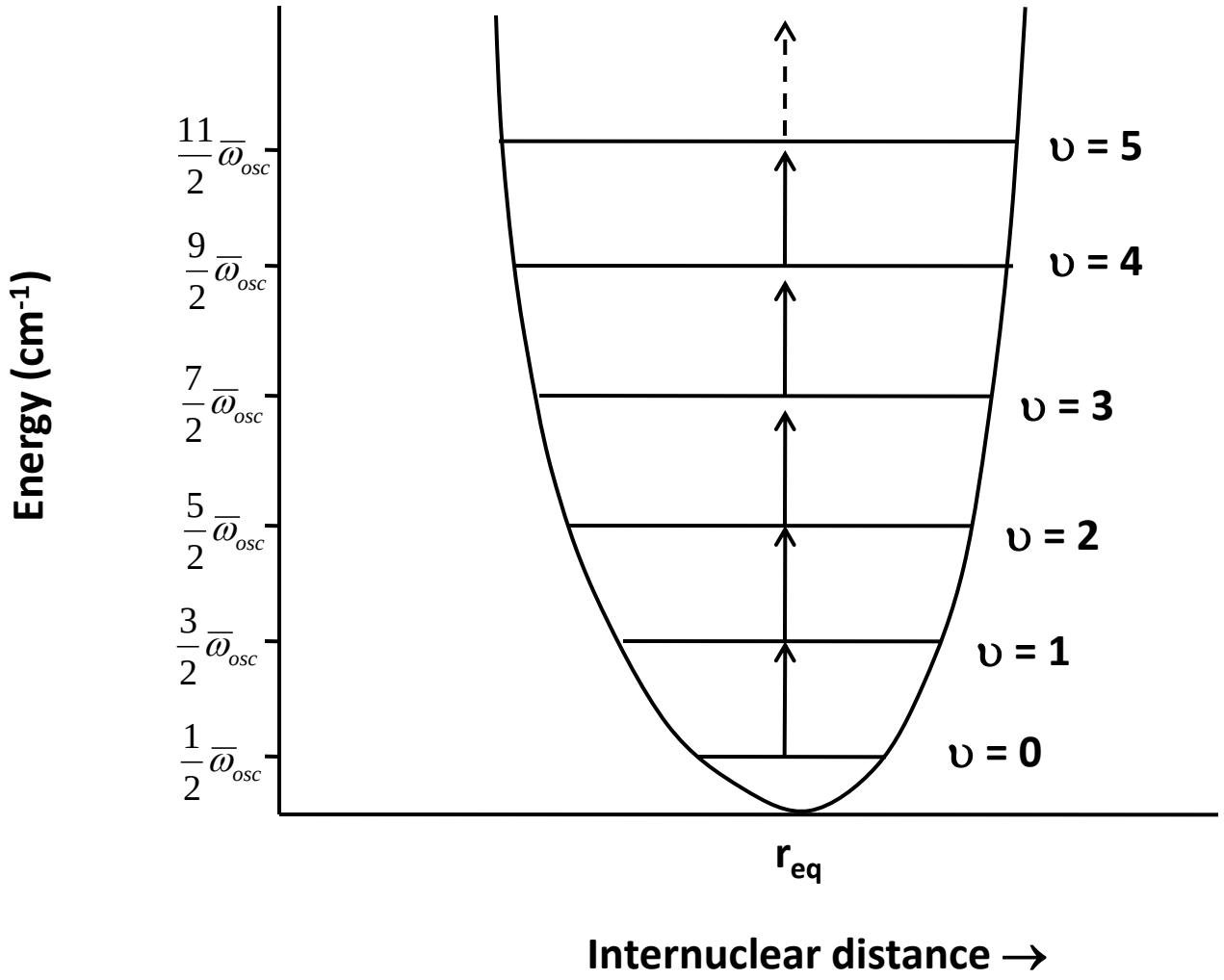
$$\varepsilon_{\upsilon+1\to\upsilon} = \left(\upsilon + 1 + \frac{1}{2}\right)\overline{\omega}_{osc} - \left(\upsilon + \frac{1}{2}\right)\overline{\omega}_{osc} = \overline{\omega}_{osc} \text{ cm}^{-1}$$

For emission

$$\varepsilon_{\upsilon\to\upsilon+1} = \overline{\omega}_{osc} \text{ cm}^{-1}$$ **For absorption**

Wave number for the spectral line absorbed or emitted

$$\nu_{spectroscopic} = \varepsilon_{\upsilon\to\upsilon+1} = \overline{\omega}_{osc} \text{ cm}^{-1}$$

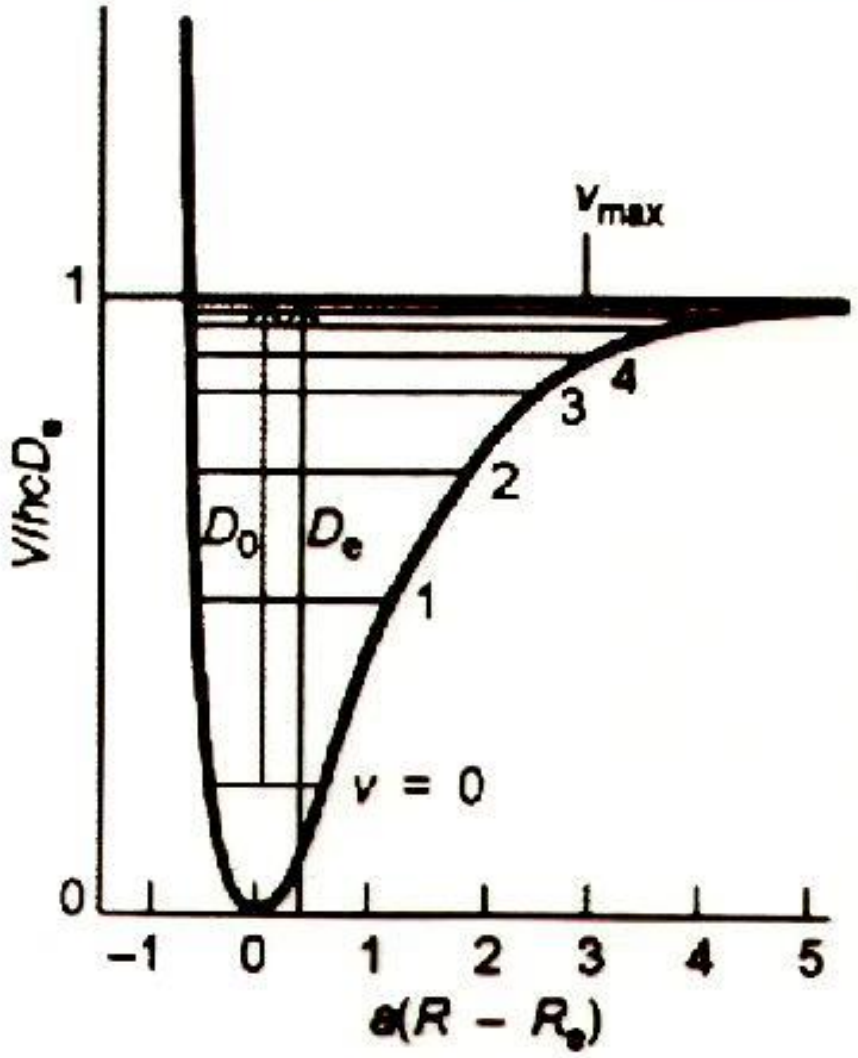

Energy (cm-1)
11/2 ω̄osc
9/2 ω̄osc
7/2 ω̄osc
5/2 ω̄osc
3/2 ω̄osc
1/2 ω̄osc
υ = 5
υ = 4
υ = 3
υ = 2
υ = 1
υ = 0
req
Internuclear distance →

As the **energy levels increase**, the particle exhibits motion that approaches **classical Newtonian mechanics**, in agreement with the **correspondence principle**.

The **correspondence principle** states that in the limit of large quantum numbers, the motion must reproduce that described by classical mechanics.

The wavelike behavior that a particle can exhibit in quantum mechanics means that it is not possible to give an exact position, only the probability of the particle occupying a spread-out region of space.

Anharmonicity



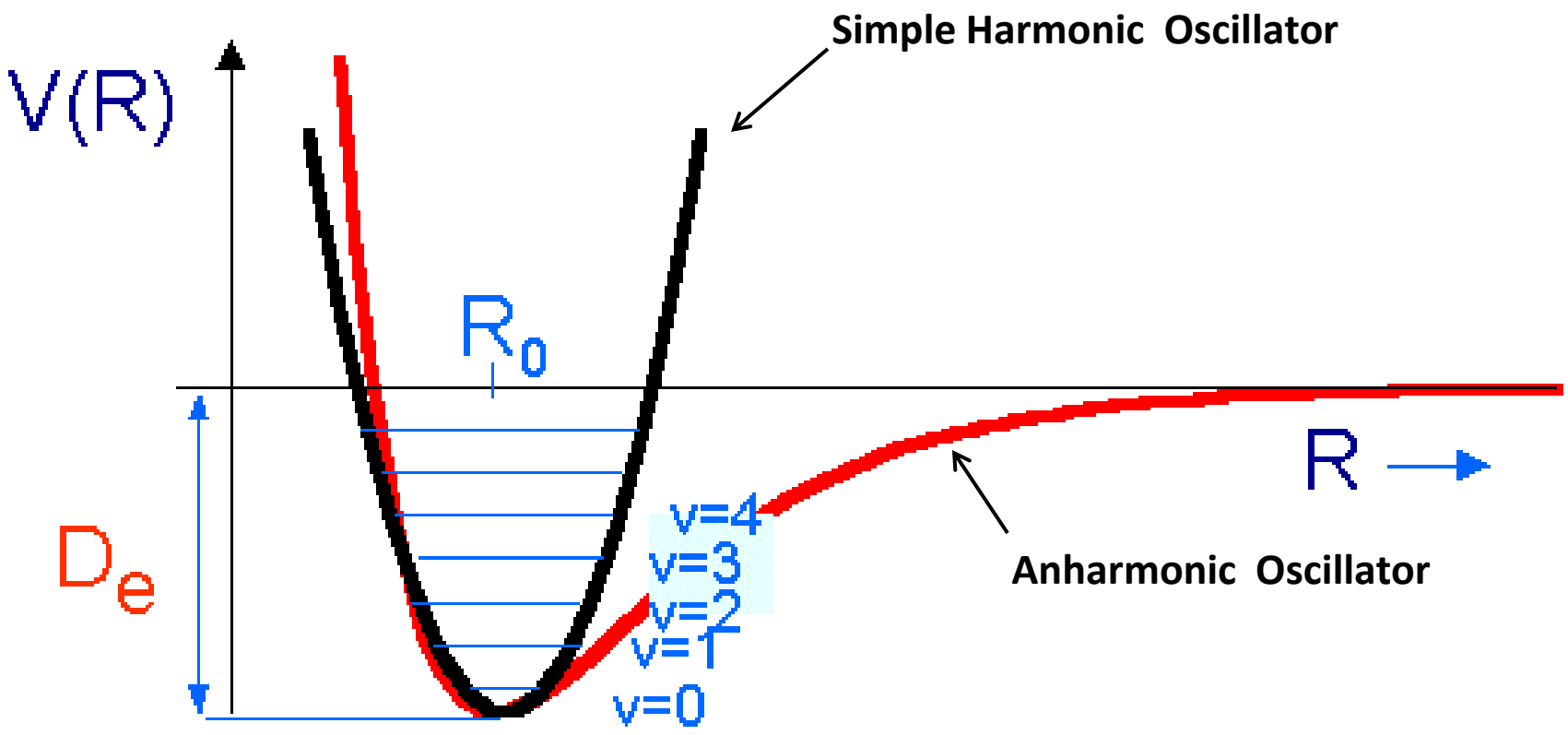
The vibrational obtained for the simple harmonic oscillator are only approximate because they are based on a parabolic approximation to the actual potential energy curve. This parabolic shape cannot be correct at all extensions of the molecule because it does not allow the molecule to dissociate into its component atoms. We can not stretch a molecular bond indefinitely, eventually it must break, the harmonic approximation does not allow for this. At higher vibrational frequencies the oscillations of the atoms allows the molecule to explore regions of the potential energy curve where the parabolic approximation breaks down and additional terms are required in the Taylor expansion of the potential V. The vibrational motion then becomes **anharmonic**, in the sense that the restoring force is no longer proportional to the displacement. Because the actual (real) potential energy curve is less confining than a parabola we can anticipate that the energy levels become less widely spaced at high excitations.

The Anharmonic Oscillator

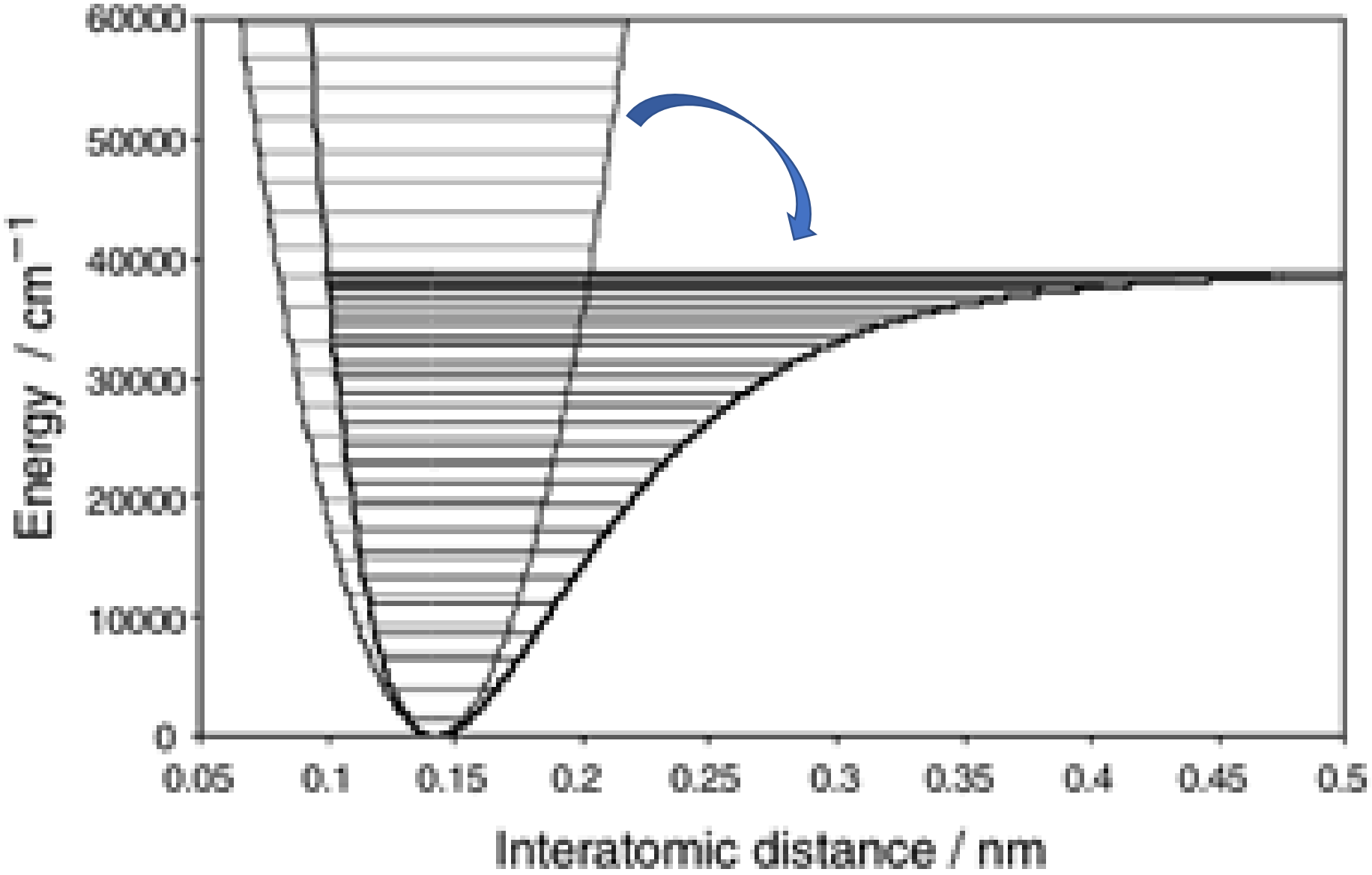
Real molecules do not obey the laws of harmonic motion.

If the bond between atoms is stretched, at a certain point it will break.

For small compression and expansion, bond may be considered as perfectly elastic
BUT
For larger amplitudes the behavior is much more complicated.


Simple Harmonic Oscillator
V(R)
R0
R
De
v=4
v=3
v=2
v=1
v=0
Anharmonic Oscillator

Vibrational energy levels

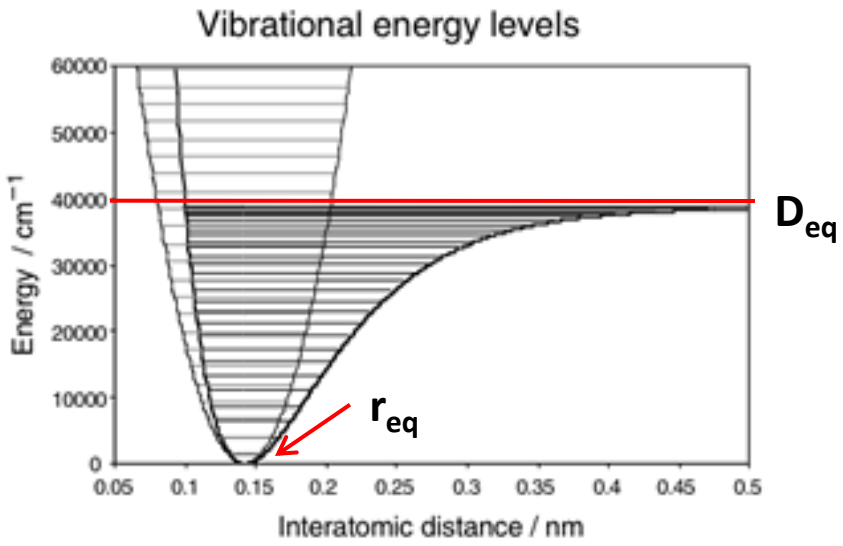


The Morse Function

A purely empirical expression.

$$E = D_{eq}\left[1-\exp\left\{a\left(r_{eq}-r\right)\right\}\right]^2$$

a is a constant for a particular molecule and D_{eq} is the dissociation energy.



The allowed vibrational levels are given by:

$$\varepsilon_{\upsilon} = \left(\upsilon + \frac{1}{2}\right)\overline{\omega}_e - \left(\upsilon + \frac{1}{2}\right)^2 \overline{\omega}_e x_e \text{ cm}^{-1}$$

$\overline{\omega}_e$ is an oscillation frequency

χ_e is the corresponding anharmonicity constant

$$\varepsilon_\upsilon = \left(\upsilon + \frac{1}{2}\right)\overline{\omega}_e - \left(\upsilon + \frac{1}{2}\right)^2 \overline{\omega}_e x_e$$

Rewriting the equation:

$$\varepsilon_\upsilon = \overline{\omega}_e \left\{1 - x_e\left(\upsilon + \frac{1}{2}\right)\right\}\left(\upsilon + \frac{1}{2}\right) \qquad (1)$$

For harmonic oscillators:

$$\varepsilon_\upsilon = \left(\upsilon + \frac{1}{2}\right)\overline{\omega}_{osc} \qquad (2)$$

Comparing (1) and (2):

$$\overline{\omega}_{osc} = \overline{\omega}_e \left\{1 - x_e\left(\upsilon + \frac{1}{2}\right)\right\}$$

$$\overline{\omega}_{osc} = \overline{\omega}_e \left\{ 1 - x_e \left(\upsilon + \frac{1}{2} \right) \right\}$$

Thus, the anharmonic oscillator behaves like the harmonic oscillator
but
with an oscillation frequency that decreases steadily with increasing υ.

In the ground state $\upsilon = 0$.

$$\overline{\omega}_0 = \overline{\omega}_e (1 - \frac{1}{2} x_e) \ \text{cm}^{-1}$$

The zero-point energy,

$$\varepsilon_0 = \frac{1}{2} \overline{\omega}_e (1 - \frac{1}{2} x_e) \ \text{cm}^{-1}$$

The zero-point energy is different from harmonic oscillator.

The Selection Rules:

$$\Delta\upsilon = \pm 1, \pm 2, \pm 3, \ldots\ldots$$

Possibility of larger jumps.

Only the lines of $\Delta\upsilon = \pm 1$, ± 2, and ± 3 have observable intensities .

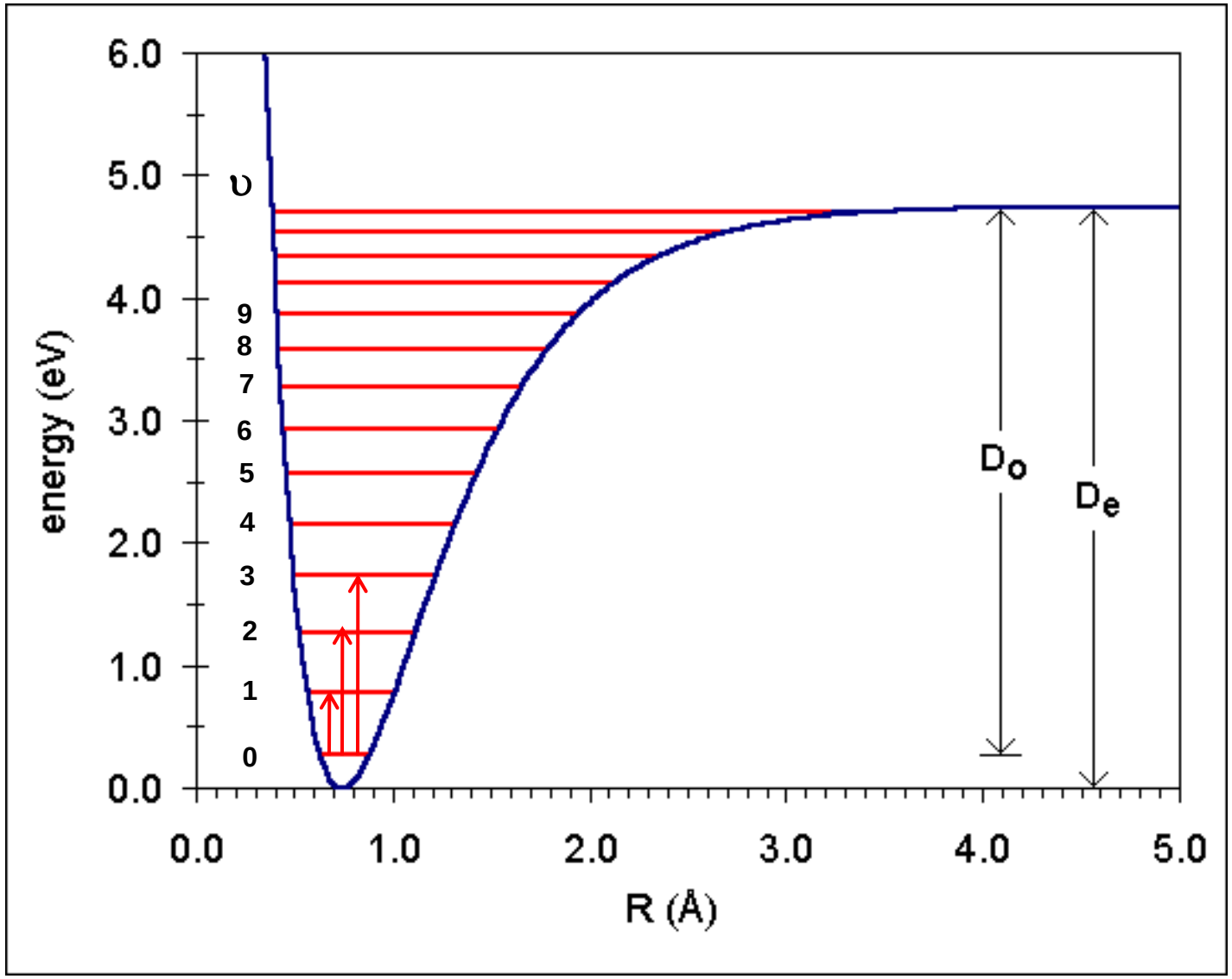
At room temperature

Spacing between vibrational levels

$$\frac{N_{\upsilon=1}}{N_{\upsilon=0}} = \exp\left\{-\frac{6.63\times10^{-34}\times3\times10^{10}\times10^{3}}{1.38\times10^{-23}\times300}\right\} \approx 0.008$$

Population of the υ = 1 state is 1% of that of the ground state.

Transitions mainly originate from υ = 0.


6.0
5.0
4.0
3.0
2.0
1.0
0.0
energy (eV)
υ
9
8
7
6
5
4
3
2
1
0
D_o
D_e
0.0
1.0
2.0
3.0
4.0
5.0
R (Å)

$\upsilon = 0 \rightarrow \upsilon = 1$, $\Delta\upsilon = +1$ (considerable intensity)

$$\Delta\varepsilon = \varepsilon_{\upsilon=1} - \varepsilon_{\upsilon=0} = \overline{\omega}_e(1 - 2x_e)\ \text{cm}^{-1}$$

$\upsilon = 0 \rightarrow \upsilon = 2$, $\Delta\upsilon = +2$ (small intensity)

$$\Delta\varepsilon = \varepsilon_{\upsilon=2} - \varepsilon_{\upsilon=0} = 2\overline{\omega}_e(1 - 3x_e)\ \text{cm}^{-1}$$

$\upsilon = 0 \rightarrow \upsilon = 3$, $\Delta\upsilon = +3$ (negligible intensity)

$$\Delta\varepsilon = \varepsilon_{\upsilon=3} - \varepsilon_{\upsilon=0} = 3\overline{\omega}_e(1 - 4x_e)\ \text{cm}^{-1}$$

The spectral lines lie very close to $\overline{\omega}_e$, $2\overline{\omega}_e$ and $3\overline{\omega}_e$

The spectral lines near $\overline{\omega}_e$ $\longrightarrow$ **Fundamental absorption**

The spectral lines near $2\overline{\omega}_e$ $\longrightarrow$ **First overtone**

The spectral lines near $3\overline{\omega}_e$ $\longrightarrow$ **Second overtone**

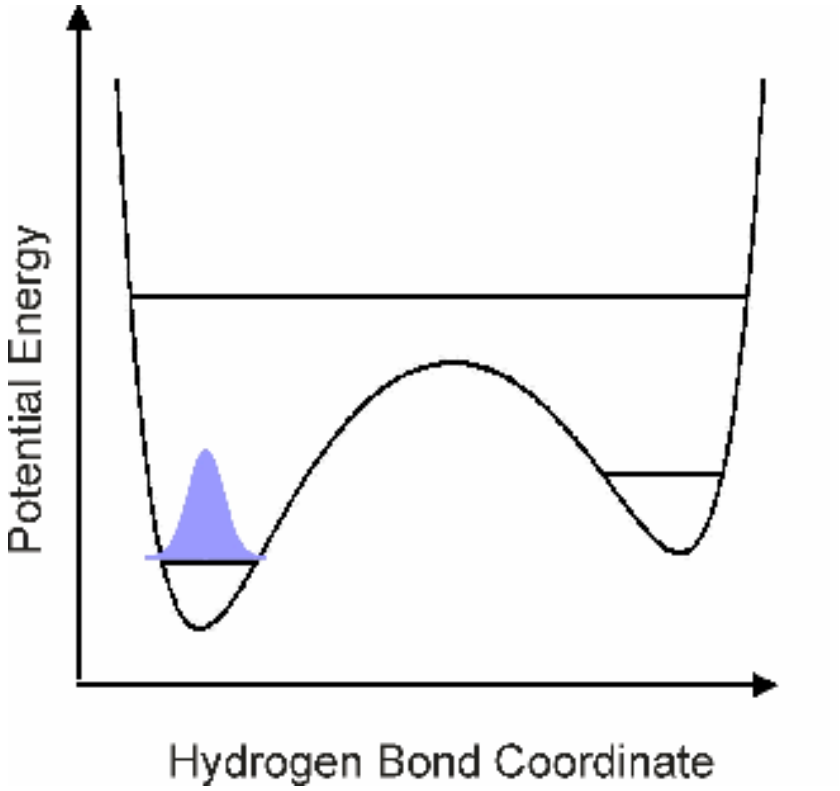
At a higher temperature the population of the υ = 1 state becomes appreciable.

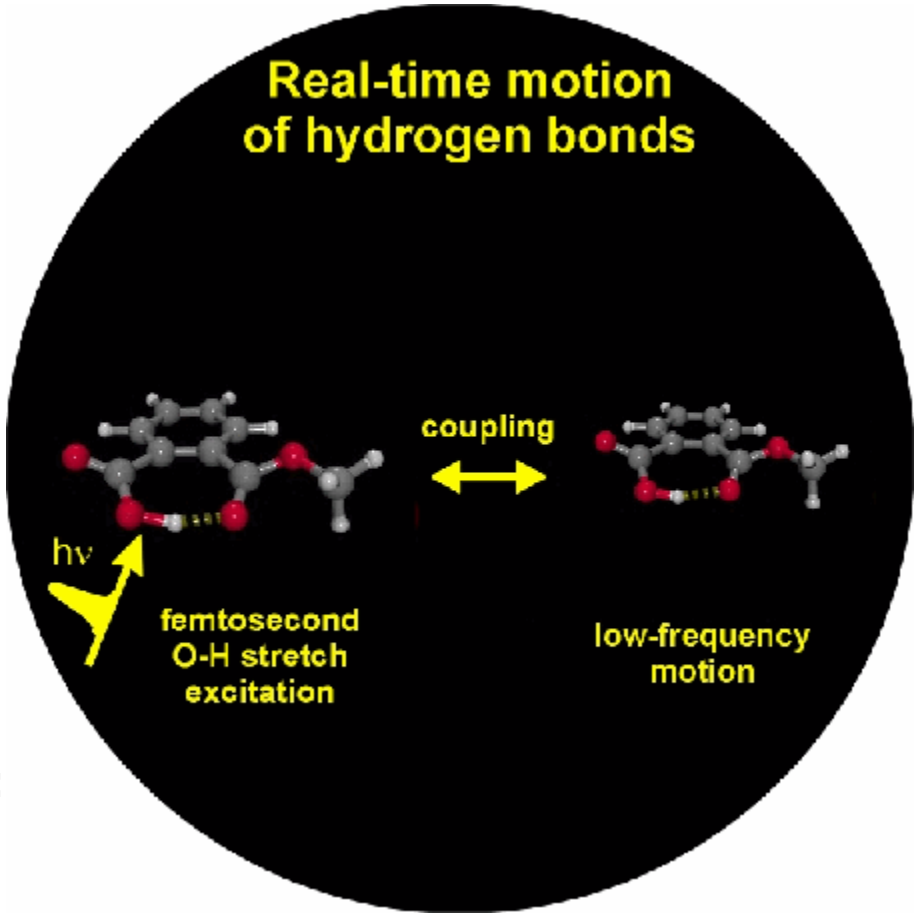
So,

$\upsilon = 1 \rightarrow \upsilon = 2$, $\Delta\upsilon = +1$ (very weak intensity)

$$\Delta\varepsilon = \varepsilon_{\upsilon=2} - \varepsilon_{\upsilon=1} = \overline{\omega}_e(1 - 4x_e)\ \text{cm}^{-1}$$

Such weak absorptions are called "**hot bands**".





Vibrational perspective of hydrogen bonds:

typically a strong anharmonic coupling exists between the O-H/O-D stretching and low-frequency modes that modulate the hydrogen bond distance.